

04-07-2025

Causal Inference w/ Unconfoundedness:

$$\rightarrow y = \alpha D + x^T \beta + e$$

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X$$

→ Decomposition 1:

$$\alpha = E[W(D, X)(L_1(Y|X) - L_0(Y|X))]$$

$$\text{s.t. } W(D, X) = 1 - L(D|X) / E[(D - L(D|X))^2]$$

$$\text{ATE} = E[E(Y|X, D=1) - E(Y|X, D=0)]$$

→ Decomposition 2:

$$\alpha = E[W(D, X)(E[Y|X, D=1] - E[Y|X, D=0]) + E[W(D, X)(E[Y|X, D=0] - L_0(Y|X))]$$

→ CATE(X)

Proof:

Add + subtract:

$$\alpha = E[W(D, X)(E[Y|X, D=1] - E[Y|X, D=0])]$$

$$- E[W(D, X)(E[Y|X, D=1] - L_1(Y|X) - E[Y|X, D=0] + L_0(Y|X))]$$

$$E[D(1 - L(D|X))E[Y|X, D=1]]$$

$$= E[(1 - L(D|X))E[Y|X, D=1] | D=1] p$$

$$= E[(1 - L(D|X))Y | D=1] p \quad (a)$$

$$E[D(1 - L(D|X))L_1(Y|X)]$$

$$= E[(1 - L(D|X))L_1(Y|X) | D=1] p$$

$$= E[(1 - L(D|X))Y | D=1] p \quad (b)$$

$$a = b \therefore E[0 - E[Y|X, D=0] - L_0(Y|X)]$$

↳ cannot get rid of second term by same argument.

□

→ Main Result on Interpreting α

Suppose unconfoundedness & overlap hold. In addition suppose either (i) $p(X) = L(D|X)$, (ii) $E[Y|X, D=0] = L_0(Y|X)$ then $\alpha = E[W(D, X)CATE(X)]$. If (i) holds, weights are ≥ 0 . Case 2 shows $E[Y|X, D=0] - L_0(Y|X) = 0$, case 1 show on hw.

→ Main result on interpreting α

- w/o T.E. homogeneity, not getting ATE but more interpretable

→ More on weights:

- $\alpha = E[D(1-L(D|X)) / E[D-L(D|X)]^2] \times CATE(X)$
 $(L|E) = E[D|X] = p(X) \Rightarrow E[p(X)(1-L(D|X)) / E[D-L(D|X)]^2] \times CATE(X)$ (approx. weighted by var.)
- $var(D) = E[D^2] - E[D]^2 = p(1-p)$ (max at $p=0.5$)
- $var(D|X) = p(X)(1-p(X))$
- when does $\alpha = ATE$?
 $CATE(X) = ATE$ (weights mean 1 and CATE comes out)

→ $ATT = E[Y(1) - Y(0) | D=1]$
 $= E[Y|X, D=1] - E[E[Y|X, D=0] | D=1]$

alt approaches: reg adjustment

- Assume linear model for untreated potential outcomes

$$E[Y|X, D=0] = L_0(Y|X) = X^T \beta_0$$

$$\Rightarrow ATT = E[Y|D=1] - E[X^T \beta_0 | D=1]$$

$$= E[Y|D=1] - E[X|D=1]^T \beta_0$$

- Step 1: est $\hat{\beta}_0$ from reg of Y on X using untreat only

- Step 2: $\hat{ATT} = \bar{Y}_{D=1} - \bar{X}_{D=1}^T \hat{\beta}_0$ (regression adjustment)

$$= \frac{1}{n} \sum_{i \in \text{treat}} (Y_i - X_i^T \hat{\beta}) \quad (\text{imputation})$$